

Pneumonia Detection Using Anchor-Free Object Detection

An FCOS-based study with ResNet backbones

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Outline

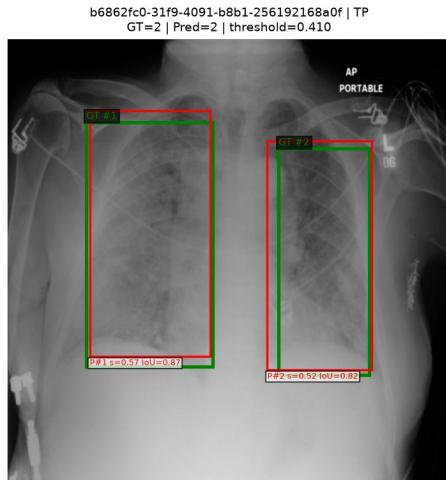
- 1 Introduction
- 2 Architecture
- 3 ResNet Backbone
- 4 Experimental Setup
- 5 Results

Motivation and Objective

- Pneumonia detection from chest X-ray images is a challenging medical imaging task.
- The objective is not only to determine whether pneumonia is present, but also to **localize the affected region**.
- Pneumonia findings can vary substantially in **size, shape and position**.
- This motivates the use of an object detection framework rather than image-level classification alone.

Main question:

How do backbone depth and pretraining affect an anchor-free pneumonia detector?



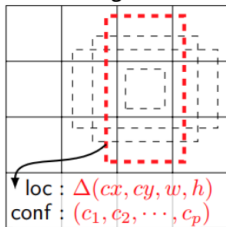
Example chest X-ray with pneumonia
localization

From Anchor-Based to Anchor-Free Detection

Anchor-Based

- Predefined bounding boxes
- Classify anchors and regress box offsets
- Requires anchor sizes and aspect ratios

Anchor based detection
using SSD



Anchor-Free

- No predefined bounding boxes
- Predictions from spatial locations
- FCOS: class + (l, t, r, b) + centerness

Anchor free detection
using FCOS



This project focuses on the anchor-free formulation.

From the Reference Study to the Project Implementation

Reference

Wu et al. (2024)

- Pneumonia detection on the RSNA dataset
- Anchor-free detection framework
- Feature pyramid
- Center and scale prediction

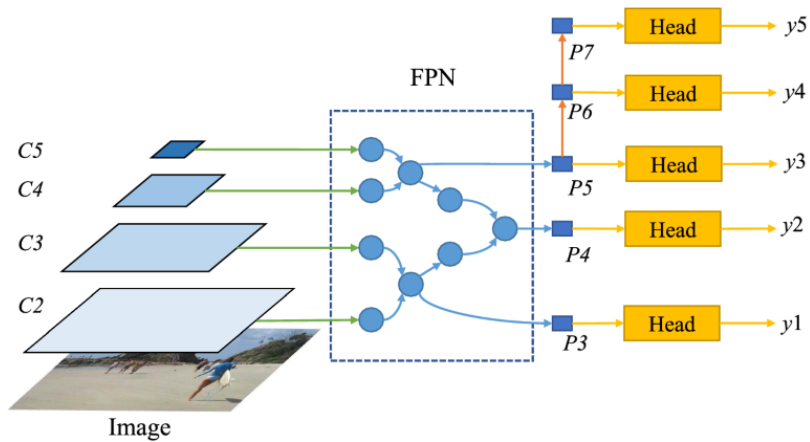
This project

FCOS-based implementation

- ResNet-50 / ResNet-101
- Feature Pyramid Network
- Classification
- LTRB regression
- Centerness

Goal: study the FCOS formulation itself rather than reproduce the reference implementation as a black box.

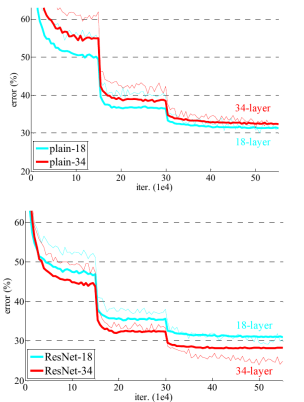
Overall Architecture



Input → ResNet → FPN → Detection Heads

Why ResNet?

- Increasing depth does not always improve training.
- He et al. observed a **degradation problem**: deeper plain networks can have **higher training error**.
- The effect is not simply due to overfitting.
- Main issue: **optimization difficulty**.



Key idea

ResNet introduces **residual learning** and shortcut connections to make deeper networks easier to optimize.

How Residual Learning Works

Plain formulation

$$H(x)$$

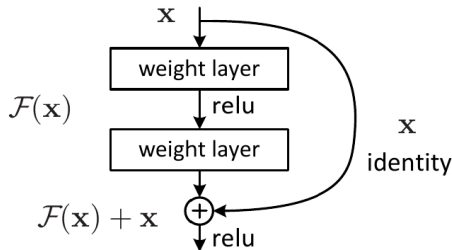
The stacked layers directly learn the desired transformation.

Residual formulation

$$F(x) = H(x) - x$$

so that

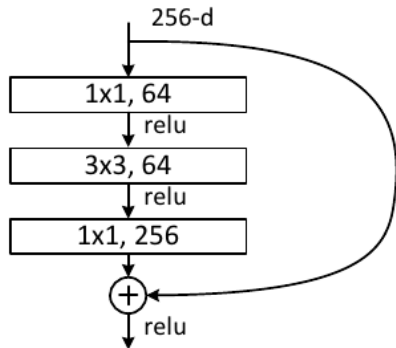
$$H(x) = F(x) + x$$



Shortcut connection + residual branch

The shortcut provides a direct identity path through the network.

The Bottleneck Residual Block



Three-layer residual branch

- 1 $\times$ 1 convolution: **reduce** channels
- 3 $\times$ 3 convolution: spatial feature extraction
- 1 $\times$ 1 convolution: **restore** channels

$$C_{\text{in}} \rightarrow C_{\text{bottleneck}} \rightarrow C_{\text{out}}$$

Why? Efficient deeper networks

For ResNet-50/101, the bottleneck uses 1×1 , 3×3 , and 1×1 convolutions.

Stage	R50	R101	Resolution	Channels
Input	—	—	512×512	3
Conv1	—	—	256×256	64
MaxPool	—	—	128×128	64
$conv2_x$	3	3	128×128	256
$conv3_x$	4	4	64×64	512
$conv4_x$	6	23	32×32	1024
$conv5_x$	3	3	16×16	2048

Conv1: 7×7 , 64 channels, stride 2

MaxPool: 3×3 , stride 2

Key idea

- Same bottleneck architecture
- R101 is deeper mainly in $conv4_x$
- Resolution decreases
- Channels increase

C_3 , C_4 , C_5

Feature maps extracted from the backbone

Why Multi-Scale Features?

Deep layers

- High-level semantic information
- Large receptive fields
- Low spatial resolution

Shallow layers

- High spatial resolution
- Precise localization information
- Weaker semantic representation

The trade-off

Semantic information $\leftrightarrow$ Spatial precision

- Using only deep features: **good semantics, poor localization**
- Using only shallow features: **good localization, weaker semantics**

Object detection needs both.

How the Feature Pyramid Network Works

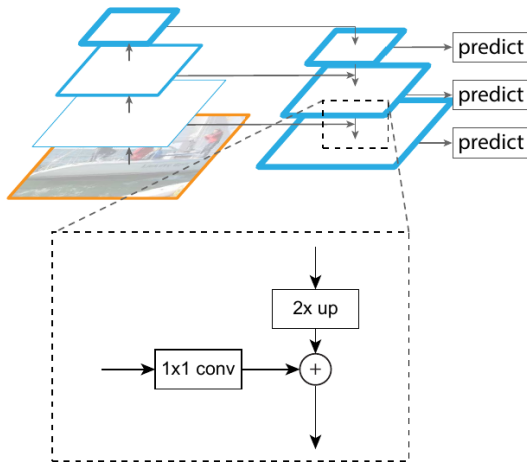
Deep layers

- Strong semantic information
- Large receptive fields
- Lower spatial resolution

Shallow layers

- Higher spatial resolution
- Better localization detail
- Weaker semantic representation

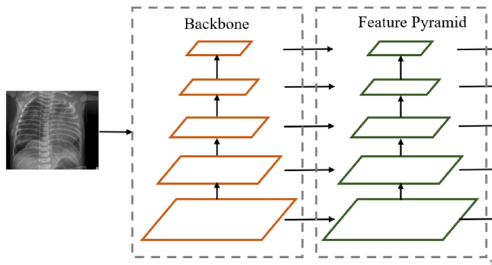
FPN combines both



Top-down pathway + lateral connections

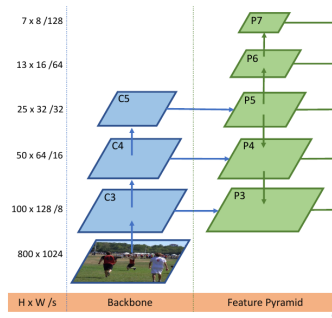
Reference Pneumonia Framework vs. FCOS

Reference pneumonia framework



Five backbone outputs are independently projected via 1×1 convolutions.

FCOS



C_3 – C_5 are fused through a top-down FPN, then extended with P_6 and P_7 .

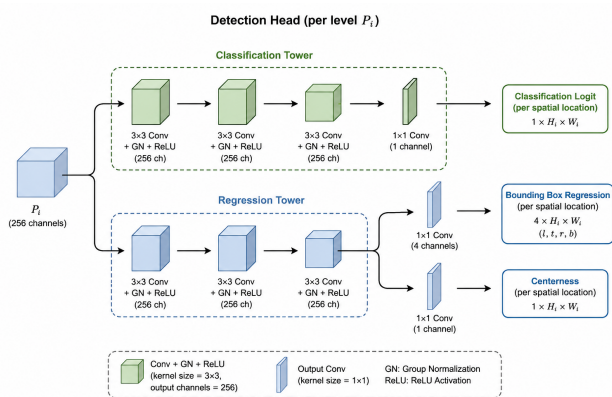
The project's implementation follows the FCOS feature construction.

FCOS-Style Detection Head

- One independent head for each FPN level P_3 – P_7
- Each input feature map has **256 channels**
- Two feature towers:
 - Classification
 - Regression

Per-location predictions

class (l, t, r, b) centerness



What Does Each Head Predict?

Classification

$$1 \times H_i \times W_i$$

Is this location associated with pneumonia?

Box Regression

$$4 \times H_i \times W_i$$

(l, t, r, b) distances

Centerness

$$1 \times H_i \times W_i$$

Quality of the spatial location

$$P_i \longrightarrow \boxed{\text{Classification}} + \boxed{\text{Regression}} + \boxed{\text{Centerness}}$$

Key point

Predictions are associated directly with spatial locations, not anchor boxes.

Target Assignment

1. Map locations to the image

For location (x, y) and stride s :

$$p_x = (x + 0.5)s, \quad p_y = (y + 0.5)s$$

A location is positive only if it lies inside a ground-truth box.

2. Build LTRB targets

$$l = p_x - x_1, \quad t = p_y - y_1$$

$$r = x_2 - p_x, \quad b = y_2 - p_y$$

3. Assign the FPN level

$$d_{\max} = \max(l, t, r, b)$$

P_3	$[0, 64)$
P_4	$[64, 128)$
P_5	$[128, 256)$
P_6	$[256, 512)$
P_7	$[512, \infty)$

4. Centerness target

$$c^* = \sqrt{\frac{\min(l, r)}{\max(l, r)} \frac{\min(t, b)}{\max(t, b)}}$$

GT boxes $\rightarrow$ positive mask + LTRB + centerness targets

Loss Functions

Classification — Sigmoid Focal Loss

$$\mathcal{L}_{\text{cls}} = -\alpha_t(1 - p_t)^\gamma \log(p_t)$$

$$\alpha = 0.25, \quad \gamma = 2$$

Bounding Box — GloU Loss

$$\text{GloU}(A, B) = \text{IoU}(A, B) - \frac{|C \setminus (A \cup B)|}{|C|}$$

$$\mathcal{L}_{\text{reg}} = 1 - \text{GloU}(A, B)$$

Centerness — Sigmoid Focal Loss

$$\mathcal{L}_{\text{ctr}} = -\alpha_t(1 - p_t)^\gamma \log(p_t)$$

Experimental Setup

Dataset

- RSNA Pneumonia Detection Challenge
- Bounding-box annotations
- Input resolution: 512×512

Common Pipeline

- Same detector architecture
- Same data split
- Same target assignment
- Same evaluation procedure

Backbone Configurations

	ImageNet	Chest-Xray
ResNet-50	R50-IN	R50-CX
ResNet-101	R101-IN	R101-CX

2 depths $\times$ 2 initializations = 4 final models

Goal: study the effect of backbone depth and initialization.

Optimization

- Backbone freezing
- Differential learning rates
- Learning-rate warm-up
- Cosine annealing

Stability

- Gradient clipping
- Exponential Moving Average (EMA)
- Validation-based checkpoint selection

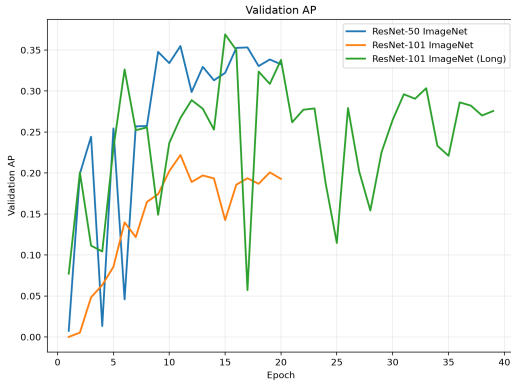
Freeze → Warm-up → Unfreeze → Fine-tune

Main idea

Protect pretrained features first,
then progressively adapt the detector.

Training schedules were adapted to the
convergence behaviour of each configuration.

Why Longer Training?



ResNet-101

Slower convergence and larger validation fluctuations

Simply increasing the number of epochs was not sufficient.

Evaluation Metrics

Detection

AP

Average Precision across confidence thresholds

AR@10

Average Recall with at most 10 detections/image

Image-level

Precision

Fraction of positive predictions that are correct

Recall

Fraction of positive images correctly detected

F1-score

Balance between precision and recall

Localization

Mean IoU

Average overlap of matched boxes

Box F1

Box-level precision–recall balance

Operating point

For image-level metrics, the confidence threshold is selected on the validation set by maximizing Youden's J .

Quantitative Results

Model	AP	AR@10	Image F1	Mean IoU	Box F1
R50-IN	0.423	0.886	0.654	0.688	0.448
R50-CX	0.399	0.876	0.633	0.684	0.424
R101-IN	0.397	0.901	0.668	0.692	0.446
R101-CX	0.239	0.804	0.649	0.659	0.388

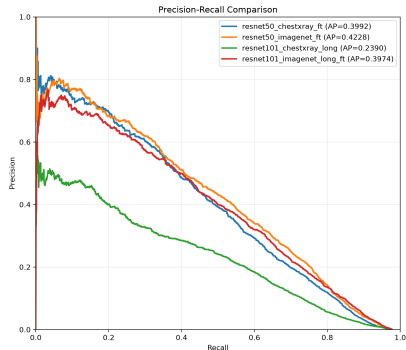
R50-IN

Highest AP and slightly highest Box F1.

R101-IN

Highest AR@10, Image F1 and mean matched IoU.

Precision–Recall and Object Size



Precision–Recall curves on the validation set

Key observation

R50-IN provides the strongest integrated Precision–Recall trade-off.

Overall AP

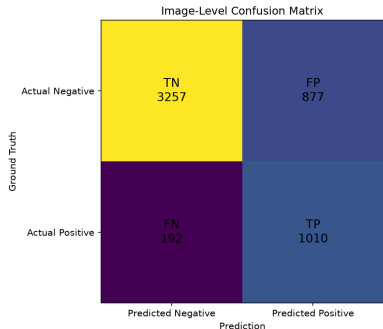
Model	AP
R50-IN	0.423
R50-CX	0.399
R101-IN	0.397
R101-CX	0.239

By region size

Model	AP _M	AP _L
R50-IN	0.039	0.442
R50-CX	0.024	0.433
R101-IN	0.035	0.413
R101-CX	0.029	0.231

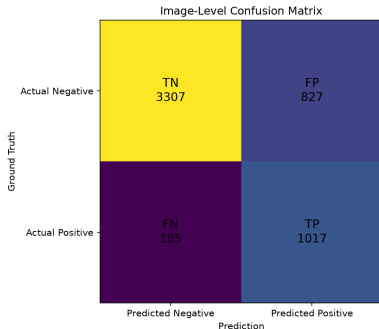
Image-Level Results: ImageNet Initialization

R50-IN



F1 = 0.654 Youden J = 0.628

R101-IN



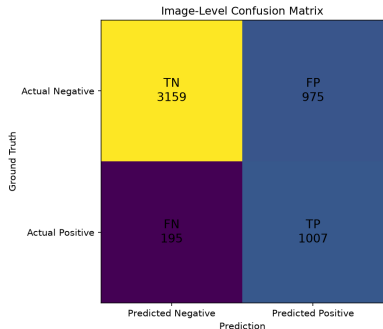
F1 = **0.668** Youden J = **0.646**

Observation

R101-IN provides the stronger image-level operating point.

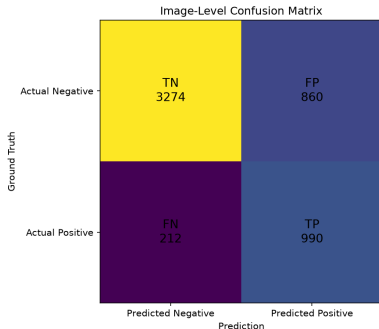
Image-Level Results: Chest-Xray Initialization

R50-CX



F1 = 0.633 Youden J = 0.602

R101-CX



F1 = 0.649 Youden J = 0.616

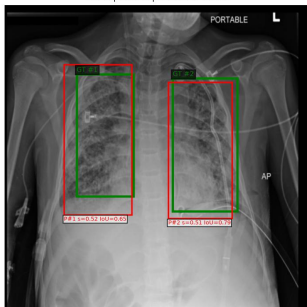
Observation

ImageNet initialization performs better than Chest-Xray initialization under the final training conditions.

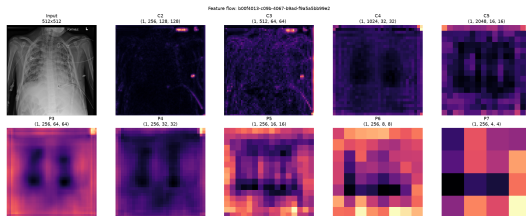
Qualitative Example: R101-IN

Input radiograph

b00f4013-c09b-4067-b9ed-f9a5a5bb99e2 | TP
GT=2 | Pred=2 | threshold=0.410



Feature flow and prediction



Qualitative inspection

The detector progressively transforms the radiograph into multi-scale feature representations before producing the final prediction.

Clinical Review: Limitations of the Dataset

Image acquisition

- Many examinations are **portable chest X-rays**
- Patients are often examined in a **supine or non-standing position**
- Positioning can change the apparent extent and distribution of opacities
- Only **one projection** is available in the dataset

Clinical assessment

- In practice, radiologists can use **frontal and lateral projections**
- Diagnosis is based on the **overall thoracic appearance**, not only on the opacity
- Cardiac size, diaphragm and costophrenic angles can provide additional information
- Clinical history and previous imaging may further change the interpretation

Main limitation

The model receives **a single radiograph and an opacity annotation**, while clinical assessment can integrate **multiple projections and additional anatomical and clinical information**.

Software

- PyTorch
- torchvision

Datasets

RSNA Pneumonia Detection Challenge

Used for detector training and evaluation.

Chest X-Ray Images (Pneumonia)

Used for Chest-Xray pretraining.

Hardware

- NVIDIA RTX 4090
- Cloud GPU rental
(<https://www.runpod.io/>)
- ~20 experiments
- Total cost: ~20

Repository

<https://github.com/Francesco2035/fcos-pneumonia-detection-naml-25-26>

Conclusions

1. Anchor-free detection

A fully convolutional FCOS-style detector can localize pneumonia without predefined anchor boxes.

2. Initialization and backbone

ImageNet initialization provided the most reliable starting point. The Chest-Xray pretrained models were relatively disadvantageous, especially for ResNet-101.

3. Main lesson

A strong backbone is important, but **careful training and optimization** have a major impact on the final performance. Better-controlled training and richer clinical information remain key directions.

Best overall operating-point model: R101-IN

References

-  L. Wu et al., *Pneumonia detection based on RSNA dataset and anchor-free deep learning detector*, Scientific Reports, 2024.
-  Z. Tian, C. Shen, H. Chen, T. He, *FCOS: Fully Convolutional One-Stage Object Detection*, ICCV, 2019.
-  K. He, X. Zhang, S. Ren, J. Sun, *Deep Residual Learning for Image Recognition*, CVPR, 2016.
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-  H. Rezatofighi et al., *Generalized Intersection over Union: A Metric and a Loss for Bounding Box Regression*, CVPR, 2019.